



Power & Ground Plane Design

PDN impedance • Plane capacitance • Stitching vias • IR drop

The silent backbone — power & ground decide signal quality

Audience: Intermediate / Advanced Engineering Students

Plane layout • Decoupling strategy • Target impedance • EMI



Why Planes Matter

Power & ground planes do MORE than carry current

Return paths

High-freq currents return beneath signal traces.

Distributed capacitance

Plane pairs form built-in low-Z bypass capacitor.

Heat spreading

Copper sinks heat away from hot ICs.

EMI shielding

Plane stack confines fields
→ less emission.

IR drop reduction

Wide planes carry mA with negligible drop.

Reference for SI

Controlled-Z traces need continuous reference.

PDN — Power Distribution Network

Z(f) of the power rail at the IC pin

- ▶ PDN = the IC's view of its power supply impedance vs frequency
- ▶ Switching IC draws transient current → if PDN $Z > 0$, voltage sags
- ▶ Target impedance $Z_{\text{target}} = \Delta V_{\text{allowed}} / \Delta I_{\text{transient}}$
- ▶ Example: $\Delta V = 50 \text{ mV}$, $\Delta I = 1 \text{ A}$ → $Z_{\text{target}} = 50 \text{ m}\Omega$ at all relevant frequencies
- ▶ PDN must be $< Z_{\text{target}}$ from DC to the bandwidth of the IC's transient currents
- ▶ DC: regulator + bulk cap; LF: tantalum / polymer; MF: ceramic; HF: plane capacitance
- ▶ Above 1 GHz: only plane capacitance helps — vias to caps are too inductive

Plane Placement in the Stackup

Where they live changes everything

- ▶ 4-layer typical: L1 signal / L2 GND / L3 PWR / L4 signal
- ▶ L2 GND under L1 signals → microstrip reference
- ▶ L3 PWR under L4 signals → power-referenced — works but suboptimal
- ▶ Better: 4-layer = L1 sig / L2 GND / L3 GND / L4 sig — two GND planes, no PWR plane
- ▶ Power distributed as a low-impedance trace on signal layer; bulk caps everywhere
- ▶ 6-layer for serious products: L1 sig / L2 GND / L3 sig / L4 PWR / L5 GND / L6 sig
- ▶ L2-L4 plane spacing tight (< 0.1 mm) → high plane capacitance for HF decoupling
- ▶ L3 sig sandwiched between two planes = stripline (best SI)

Stitching Vias — Connecting Plane Pieces

Without them, planes are not really planes

- ▶ Stitching via = via that connects same-net plane pieces (GND-to-GND, etc.)
- ▶ When a high-speed signal changes layer, its return current must follow
- ▶ Place stitching vias adjacent to signal vias (within 1-3 mm)
- ▶ Stitching via spacing on plane: $< \lambda/20$ of highest frequency
- ▶ At 5 GHz: $\lambda/20 \approx 3$ mm spacing (FR-4)
- ▶ Stitch around the perimeter of GND pours to prevent slot-antenna effect
- ▶ Also: stitch GND across plane splits if the split is unavoidable (with caps)
- ▶ Stitching cost = free in fab (they're just standard vias)

Decoupling Cap Strategy

Multi-value caps to cover the spectrum

Cap value	Type	Effective freq	Where
100 nF (X7R 0402)	MLCC	10 MHz - 1 GHz	At every IC power pin
10 nF (X7R 0402)	MLCC	100 MHz - 1+ GHz	At every IC power pin (paired with 100n)
1 nF (X7R 0402)	MLCC	1 GHz+	For HF-sensitive analog / SerDes
10 µF (X7R 0805)	MLCC	10 kHz - 10 MHz	Per IC power region
22-47 µF (X7R 1210)	MLCC bulk	1 kHz - 1 MHz	Per voltage rail (1-2 per board)
100-470 µF	Polymer tantalum	100 Hz - 100 kHz	Power regulator output
1000+ µF	Aluminium electrolytic	< 100 Hz	Power-supply input

Placement priority

Smaller caps CLOSER to the IC. Larger caps fan out. 100 nF + 10 nF within 1-2 mm of every power pin = mandatory.

Plane-Pair Capacitance

Built-in distributed cap

- ▶ Two parallel planes form a giant capacitor: $C = \epsilon_0 \times \epsilon_r \times A / d$
- ▶ FR-4 with 0.1 mm spacing $\rightarrow$ ~ 400 pF/in² of plane area
- ▶ 100 cm² of plane area = ~ 62 nF — distributed, with NO inductance from leads
- ▶ Most effective at high frequencies (where lead inductance of discrete caps dominates)
- ▶ Achievable: very tight plane spacing (≤ 50 μ m) $\rightarrow$ 4 $\times$ more capacitance
- ▶ Tradeoff: tight planes need careful fab — increases cost
- ▶ High-speed boards (PCIe Gen 4+) often use dedicated 'embedded capacitance' material
- ▶ ER-rich materials (e.g., Faradflex, BC2000): 10-100 $\times$ plane capacitance

Plane Splits — When and Why

Splits sometimes necessary, often a mistake

Splits: pros vs cons

Reasons to split a plane

- ▶ Different voltage rails (5 V analog vs 3.3 V digital)
- ▶ AGND vs DGND (legacy mixed-signal pattern)
- ▶ Isolated power rails (e.g., relay supply)
- ▶ Power-supply input section (rectified mains)
- ▶ Modular sub-circuits with separate ground reference

Why splits hurt

- ▶ Return currents detour around split → EMI
- ▶ Controlled-Z traces crossing a split = impedance jump
- ▶ Modern best practice: single solid GND, smart placement
- ▶ Modern ADCs (Bogatin) explicitly recommend SINGLE solid ground
- ▶ If split must exist: stitch ground with caps across the split

Power Plane vs Power Trace

When dedicated plane, when trace

- ▶ Dedicated power PLANE: best for high-current rails (> 1 A continuous)
- ▶ Plane gives ultra-low impedance + heat spreading + free filtering
- ▶ Power TRACE: fine for low-current rails (< 500 mA)
- ▶ Trace cost: needs careful width sizing per current rating
- ▶ Hybrid: trace on signal layer + multiple bulk caps in parallel = decent for medium current
- ▶ For multi-voltage products: each high-current rail gets its own plane (or large pour)
- ▶ Don't waste a whole layer on a single 50 mA rail — use a trace + cap network
- ▶ 4-layer with 2-3 rails: plane = main rail; traces = others

IR Drop on Planes

Even copper has resistance

- ▶ Plane sheet resistance: 1 oz Cu $\approx$ 0.49 m Ω per square
- ▶ Long, narrow plane sections have significant resistance
- ▶ Example: 1 cm $\times$ 10 cm plane carrying 5 A $\rightarrow$ $\sim$ 5 m Ω $\rightarrow$ 25 mV drop
- ▶ Critical for low-voltage rails (1.2 V VDDQ $\rightarrow$ 25 mV is 2%)
- ▶ Solution: short + wide plane paths; multiple bypass cap locations
- ▶ Use thicker copper (2 oz) for high-current rails
- ▶ Tool: power-integrity simulator (Sigridy, HyperLynx PI) to map IR drop on board
- ▶ Probe IR drop on prototype with sense-line testing or thermal camera

How Planes Contribute to EMC

Planes are the first EMI defense

- ▶ Continuous reference plane → small loop area for signal current → less radiation
- ▶ Multi-layer stack with closely-spaced planes confines fields between them
- ▶ Stitching vias around board perimeter prevent slot antenna effect
- ▶ Power planes act as bypass cap → less switching noise on supply
- ▶ EBG (Electromagnetic Bandgap) structures cut specific resonances (advanced)
- ▶ Decoupling capacitor placement affects EMC strongly — close = better
- ▶ Star vs uniform plane impedance pattern → cavity resonance frequencies
- ▶ Bad layout = textbook EMC compliance failure; good layout = effortless pass

Plane Discontinuities to Avoid

What breaks the magic

- ▶ Holes through planes (vias, mounting holes) — minor, OK if scattered
- ▶ Plane splits — major; force return current detour
- ▶ Cutouts under high-density components — local impedance jump
- ▶ Slot antenna (long thin gap in plane) — radiates at specific frequencies
- ▶ Plane width abruptly changing — partial reflection
- ▶ Plane voids in escape routing under BGA — return current absent
- ▶ Mounting hole anti-pads — small but noticeable for HF signals nearby

PDN Design Tools

Don't guess at impedance

Sigrity PowerSI

Cadence — full-board PDN sim
+ IR drop mapping

HyperLynx PI

Mentor — similar to PowerSI
Cheaper licence

Ansys SIwave

Field solver + PI
Industry-standard

Faraday/Bogatin tools

Free + paid PDN calculators
Good for sanity checks

Cap selection apps

Murata Solver, Yageo, Samsung
Free, online

KiCad

Basic plane visualization
No full PDN sim built-in

Common Plane Mistakes

What goes wrong in real designs

- ▶ Single-value decoupling (only 100 nF) → fails above 100 MHz
- ▶ Decoupling cap with via stub → cap inductance dominates → useless above 100 MHz
- ▶ Plane split that high-speed signal must cross → SI + EMI fail
- ▶ Power plane too narrow for current → IR drop + heating
- ▶ Not stitching ground around perimeter → slot antenna
- ▶ Long narrow ground pour (vs solid plane) → return current must detour
- ▶ Missing stitching vias near layer-change vias → return path discontinuity
- ▶ Bulk cap (47 μ F) placed at corner of board, far from IC → useless

Key Takeaways

Key takeaways from this lecture

Plane = reference + return

Both directions: forward + return.

PDN < Z_target everywhere

DC to BW of IC's transients.

Multi-value decoupling

100 nF + 10 nF + 1 nF + bulk.

Solid plane > split

Modern best practice.

Stitching vias = free

Use them generously.

Simulate when serious

PI tools for boards > 1 A or > 100 MHz.

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- ▶ Eric Bogatin — 'Signal & Power Integrity Simplified'
- ▶ Larry Smith — 'Power Integrity for Digital Designers'
- ▶ Murata MLCC simulation tool — model decoupling cap behavior
- ▶ Sigrity PowerSI / HyperLynx PI — commercial PI tools
- ▶ Ansys SIwave — combined SI + PI field solver
- ▶ Robert Feranec YouTube — PDN walkthrough videos
- ▶ DesignCon papers on PDN target impedance
- ▶ Henry Ott — EMC chapter on grounding strategies

Thank you • Build something this week.